

DFA-Net: An Efficient Diffusion With Directional Frequency Feature Augmentation for Low-Light Image Enhancement

Supplementary materials

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S.I. NETWORK STRUCTURE

A. DTCWT Decomposition and Reconstruction Process

As illustrated in Fig. S1, DTCWT decomposes the input signal through two parallel complex-valued wavelet trees. At each decomposition level, the signal is processed by paired low- and high-pass filter banks followed by downsampling, producing one low-frequency subband and six directional high-frequency subbands. The inverse DTCWT reconstructs the signal through corresponding inverse filtering and upsampling operations. Benefiting from its critically sampled and separable filter design, the DTCWT incurs low computational cost while providing approximate shift invariance. Consequently, the DTCWT can capture directional features from high-frequency subbands at $\pm 15^\circ$, $\pm 45^\circ$, and $\pm 75^\circ$, providing an effective basis for directional degradation modeling in low-light scenarios. Moreover, the DTCWT and its inverse are implemented using fixed filter banks without introducing additional learnable parameters. This favorable trade-off between efficiency and directional expressiveness makes the DTCWT particularly suitable for lightweight low-light enhancement frameworks.

B. Details of diffusion refinement

Forward diffusion. During training, Gaussian noise is directly added to the clean high-frequency feature following the standard forward diffusion process:

$$\hat{\mathcal{F}}_t^h = \sqrt{\bar{\alpha}_t} \hat{\mathcal{F}}_0^h + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, \quad (\text{S1})$$

where $\hat{\mathcal{F}}_0^h$ denotes the clean high-frequency feature, $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$, and $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ denotes Gaussian noise. Since low-frequency components mainly contain illumination information and global structures, diffusion refinement is performed only on high-frequency features to reduce computational overhead while preserving structural details.

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Reverse diffusion. The reverse denoising process is parameterized by predicting the injected noise ϵ_t and computing the conditional mean accordingly:

$$\mu_\theta(\hat{\mathcal{F}}_t^h, M, t) = \frac{1}{\sqrt{\alpha_t}} \left(\hat{\mathcal{F}}_t^h - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\hat{\mathcal{F}}_t^h, M, t) \right), \quad (\text{S2})$$

where $\epsilon_\theta(\cdot)$ denotes the noise prediction network conditioned on the fused attention map M . During inference, DDIM sampling is adopted to accelerate the reverse diffusion process. By removing the stochastic variance term, DDIM enables deterministic recovery of high-frequency components with fewer sampling steps. The diffusion schedule and sampling settings are provided in the implementation details.

S.II. EXPERIMENTS

A. Visualization Comparison on unpaired MEF, LIME, and NPE datasets

Fig. S2 presents a qualitative comparison on MEF[S1], LIME[S2], and NPE[S3] datasets between DFA-Net and current state-of-the-art LLIEs. As shown, KinD[S4] and Zero-DCE[S5] produce limited brightness enhancement. Retinexformer[S6] introduces noticeable noise amplification and incomplete texture recovery. DiffLL[S7] and PyDiff[S8] tend to oversmooth textures due to insufficient high-frequency restoration. SNR[S9] and URetinex-Net++[S10] exhibit contrast imbalance and inaccurate illumination adjustment. Although HVI[S11] generates comparatively natural-looking results, it often introduces over-enhancement artifacts, particularly in bright regions such as the sky, *e.g.*, on the NPE dataset. In contrast, DFA-Net produces sharper details and more natural illumination across different datasets, further validating its cross-domain generalization capability.

B. Ablation on the position of ACFG in the diffusion refinement

Table SI presents an ablation study on the injection position of ACFG within the U-Net-based denoising network. Different injection strategies lead to different trade-offs between restoration quality and computational cost. Although all-stage ACFG achieves slightly higher PSNR, it introduces considerable computational overhead with increased parameters, FLOPs, and inference time. Moreover, its lower SSIM suggests that excessive frequency guidance may introduce redundant interactions and affect structural consistency. In contrast, injecting

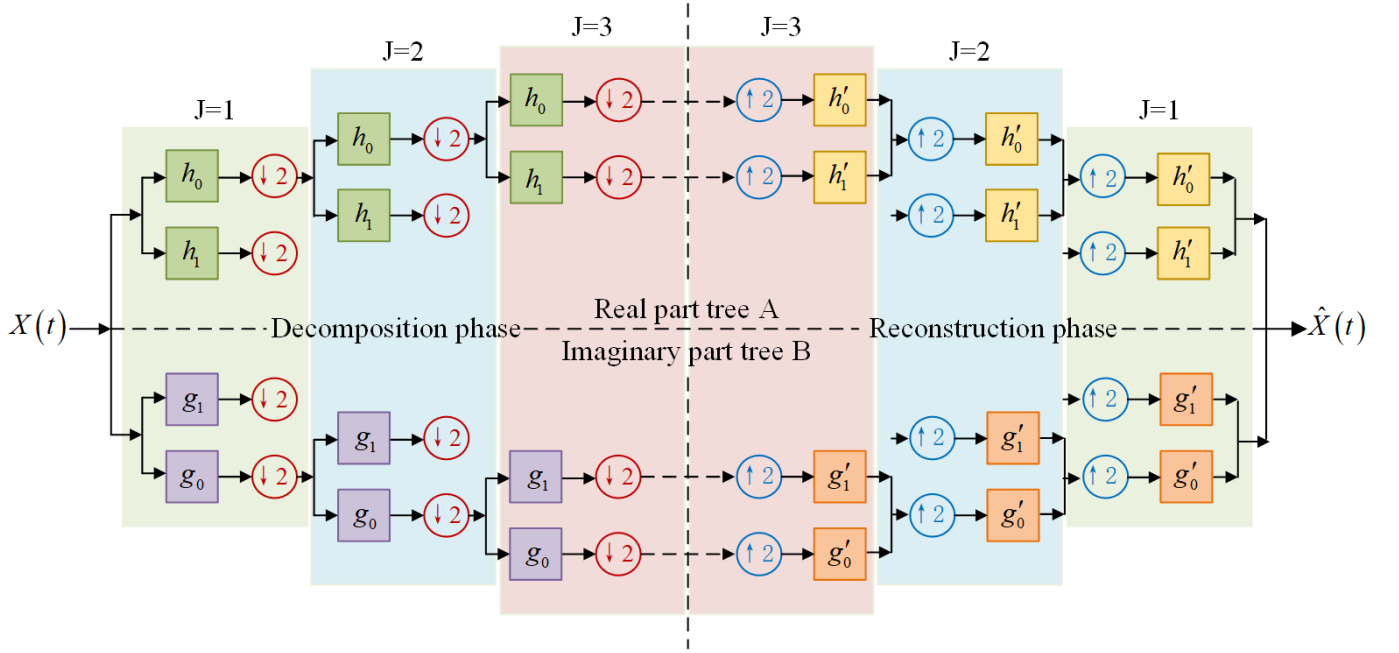


Fig. S1. The DTCWT decomposes the input signal $X(t)$ through two parallel filter trees, which apply paired filter banks $\{h_0, h_1\}$ and $\{g_0, g_1\}$ to form the real and imaginary components of complex wavelet coefficients.

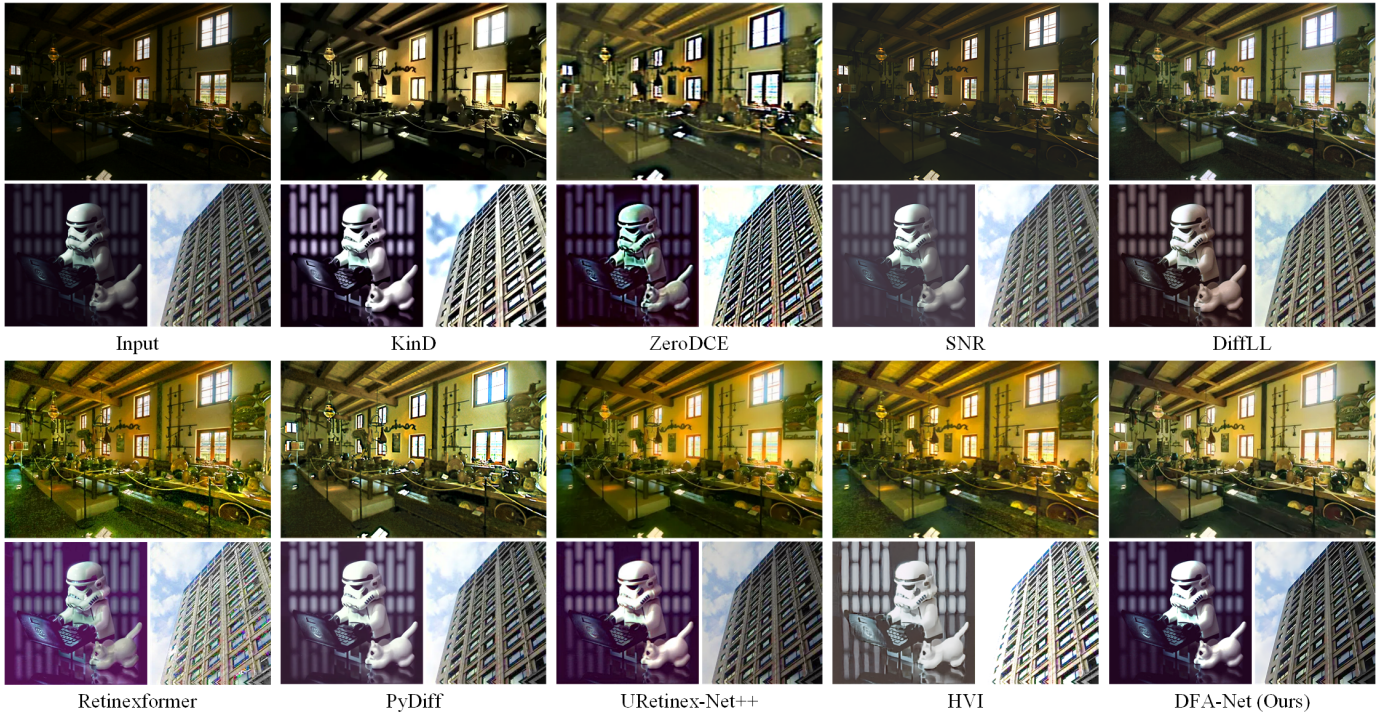


Fig. S2. Visual comparison of DFA-Net and competing methods on the unpaired MEF, LIME, and NPE datasets. The top row shows results on the MEF dataset, while the bottom row presents results on the LIME and NPE datasets.

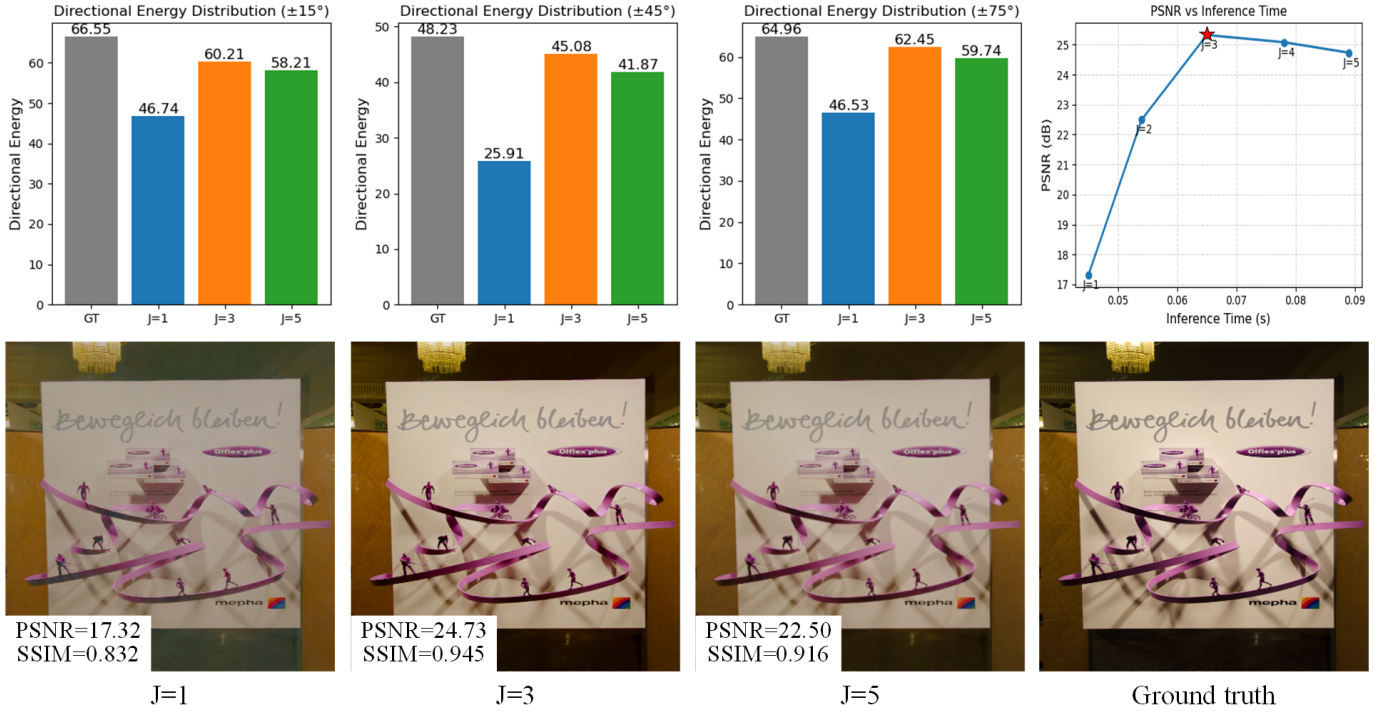


Fig. S3. The impact of different decomposition levels on the performance of DFA-Net.

TABLE SI

ABLATION STUDY ON THE INTEGRATION POSITION OF ACFG IN THE DIFFUSION DENOISING NETWORK.

ACFG Position	PSNR $\uparrow$	SSIM $\uparrow$	Params/M	FLOPs/G	Inference time (s/img)
all stages	25.27	0.901	6.17	5.61	0.228
stage-1	24.96	0.905	4.94	2.38	0.058
stage-2	25.05	0.914	5.02	2.86	0.063
stage-3 (Ours)	25.23	0.931	5.26	3.37	0.065
-	24.81	0.905	4.77	2.13	0.057

ACFG at the final upsampling stage achieves the highest SSIM and competitive PSNR with significantly lower computational cost. Therefore, the final-stage injection strategy is adopted in DFA-Net.

C. Ablation analysis of MDFA configurations

To determine the optimal configuration of MDFA, we conduct ablation studies on the number of MDFA blocks, feature dimensions, and state dimension L . The experimental results (in Table SII) demonstrate that stacking 2 MDFA blocks achieves the highest PSNR and SSIM on LOLv1, while providing the highest SSIM with slightly lower PSNR than the 3-block configuration on LOLv2-Real dataset. Further ablation on feature dimensions indicates that a channel dimension of 128 provides the best balance between restoration quality and efficiency. Regarding the state dimension L , LOLv1 yields the highest metrics at $L = 16$. For the LOLv2-Real dataset, $L = 64$ corresponds to the highest PSNR, whereas $L = 32$ corresponds to the highest SSIM. Considering computational overhead and performance gains, experiments uniformly adopt $L = 16$.

TABLE SII

ABLATION STUDY ON THE NUMBER OF MDFA BLOCKS, MODEL DIMENSIONS, AND STATE DIMENSION L ON THE LOLV1 AND LOLV2-REAL DATASET. THE OPTIMAL RESULTS OF EACH GROUP ARE MARKED IN RED.

Setting		LOLv1		LOLv2-Real	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Number of blocks	1	24.10	0.844	21.33	0.857
	2	26.38	0.935	23.56	0.904
	3	26.33	0.893	24.03	0.891
Feature dimension	32	22.96	0.811	21.65	0.864
	64	23.51	0.851	23.65	0.887
	128	25.61	0.863	23.75	0.889
	256	24.69	0.840	23.19	0.878
State dimension L	4	23.01	0.842	23.45	0.803
	8	24.57	0.863	24.96	0.812
	16	25.84	0.945	25.08	0.855
	32	25.32	0.896	25.06	0.899
	64	25.11	0.867	25.31	0.876

D. The impact of decomposition levels in DTCWT on the performance of DFA-Net

Fig. S3 illustrates the impact of different decomposition levels on the performance of DFA-Net. In the top row, the first three columns compare the directional energy distributions at $\pm 15^\circ$, $\pm 45^\circ$, and $\pm 75^\circ$ across different decomposition levels against the ground truth, highlighting the preservation of directional features at different decomposition levels. The fourth column reports the trade-off between restoration quality and efficiency by plotting PSNR against inference time for varying decomposition levels. The bottom row presents qualitative comparisons and the corresponding quantitative metrics for different decomposition levels. The results indicate that setting $J = 3$ yields the optimal visual quality and quantita-

tive metrics. Moreover, the directional energy distributions at $J = 3$ exhibit the greatest consistency with the ground truth. Importantly, this setting also achieves an effective balance between inference time and PSNR. Since directional frequency energy reflects the preservation of structural responses, higher consistency with GT indicates better recovery of degraded directional frequency features.

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